

NTN-O-RAN Platform with RL-Based Power Control and ASN.1-Optimized E2 Interface

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Abstract—Non-Terrestrial Networks (NTN) are critical for global 6G coverage, yet current O-RAN platforms lack comprehensive NTN support. We present an NTN-O-RAN platform integrating OpenNTN channel models with O-RAN E2 interface extensions, featuring three key contributions: (1) a novel E2SM-NTN service model with 33 NTN-specific KPMs enabling intelligent RAN control for satellite networks, (2) ASN.1 PER encoding achieving 93% message size reduction (1,359 bytes to 92 bytes), and (3) reinforcement learning-based power control using DQN and PPO agents that achieve approximately $5\times$ fewer RSRP violations compared to a rule-based baseline (3.48% and 4.52% vs. 18.63–23.25%), at the cost of ~ 6 dB higher transmit power. Both RL results are highly statistically significant (DQN: $t = 42.06$, $p < 10^{-64}$; PPO: $t = 51.77$, $p < 10^{-117}$). Our open-source implementation provides production-ready Docker containers and serves as a standardization candidate for O-RAN NTN extensions.

Index Terms—Non-Terrestrial Networks, O-RAN, E2 Interface, LEO Satellites, Reinforcement Learning, DQN, PPO, ASN.1 Encoding, 6G

I. INTRODUCTION

The evolution toward 6G wireless networks demands ubiquitous global connectivity, extending beyond terrestrial coverage to integrate satellite, aerial, and maritime communications [1], [2]. Non-Terrestrial Networks (NTN), particularly Low Earth Orbit (LEO) satellite constellations like Starlink (8,805 satellites) and OneWeb (588 satellites), offer unprecedented opportunities for global broadband access. However, integrating NTN into Open Radio Access Network (O-RAN) architectures presents unique challenges absent in terrestrial deployments.

A. Motivation and Problem Statement

Current O-RAN platforms, designed for terrestrial base stations, inadequately support NTN-specific characteristics:

- **High Doppler Shift:** LEO satellites at 550km altitude moving at 7.5 km/s induce ± 45 kHz Doppler shifts at S-band (2 GHz), requiring continuous frequency pre-compensation.
- **Long Propagation Delays:** Round-trip times of 11–25ms (LEO) to 540ms (GEO) violate terrestrial timing assumptions and require protocol adaptations.
- **Frequent Handovers:** Satellite visibility windows of 5–10 minutes necessitate handovers every 3–7 minutes, $10\times$ more frequent than terrestrial scenarios.
- **Dynamic Link Budgets:** Slant range variations (800–2,000 km for LEO) cause 15–20 dB path loss fluctuations, plus 0–40 dB rain attenuation.

Existing solutions fall short: ns-3 simulations [3] lack real-time O-RAN integration, commercial systems remain proprietary, and O-RAN specifications [4], [5] provide no NTN-specific service models. The gap between NTN capabilities and O-RAN intelligence frameworks limits deployment of intelligent RAN control for satellite networks.

B. Our Approach and Contributions

We present an NTN-O-RAN platform with comprehensive integration of satellite channel modeling, orbital mechanics, O-RAN E2 interface extensions, and RL-based power control. Our system architecture (Fig. 1) combines:

- **OpenNTN Integration:** 3GPP TR 38.811-compliant channel models for LEO/MEO/GEO with NVIDIA Sionna GPU acceleration
- **SGP4 Orbit Propagation:** Real-time tracking of 8,805 Starlink satellites with < 0.5 km accuracy
- **E2SM-NTN Service Model:** Novel O-RAN service model (RAN Function ID 10) with 33 NTN-specific KPMs
- **Predictive xApps:** 60-second advance handover prediction and weather-aware power control
- **ITU-R P.618 Weather Integration:** Real-time rain attenuation modeling for link adaptation

Novel Contributions:

- 1) **First O-RAN NTN platform with RL-based power control:** Integrates OpenNTN + Sionna with O-RAN E2 interface and DQN/PPO agents for intelligent power control.
- 2) **E2SM-NTN Service Model:** Novel service model with 33 NTN-specific KPMs (elevation angle, Doppler shift, propagation delay, handover prediction), 6 event triggers, and 6 control actions. Enables near-RT RIC to perform NTN-aware decisions.
- 3) **ASN.1 PER Optimization:** Achieves 93.2% message size reduction ($1,359 \rightarrow 92$ bytes) through domain-specific encoding, reducing E2 interface bandwidth requirements by $15\times$.
- 4) **Predictive Handover Architecture:** SGP4-based orbital prediction provides 60-second advance warning, enabling preparation-based handover (vs. emergency reactive handover).
- 5) **Statistical Validation:** Rigorous comparison of DQN and PPO agents against rule-based baseline. Both agents achieve $\sim 5\times$ fewer RSRP violations with high statistical significance ($p < 10^{-64}$).

C. Paper Organization

Section II surveys related work in NTN standards, O-RAN architecture, and simulation tools. Section III details system design including E2SM-NTN specification and predictive handover. Section IV describes implementation with technology stack and performance optimizations. Section V presents comprehensive experimental results with statistical validation. Section VI concludes with impact assessment and future directions.

II. RELATED WORK

A. 3GPP NTN Standardization

3GPP has progressively addressed NTN integration across multiple releases. Release 17 (TR 38.821 [2]) introduced NTN solutions for NR, defining scenarios for LEO (600-1200 km), MEO (7,000-25,000 km), and GEO (35,786 km) satellites. Key technical adaptations include timing advance adjustments for propagation delays, frequency pre-compensation for Doppler shifts, and HARQ modifications for long RTT.

Release 18 (TR 38.863 [6]) advanced regenerative payload architectures where satellites host full gNB functionality, enabling inter-satellite links and on-board beamforming. Release 19 continues with NTN enhancements for mobility, handover optimization, and IoT support [7].

However, 3GPP specifications focus on air interface and protocol adaptations, lacking RAN intelligence integration. Our work bridges this gap by extending O-RAN E2 interface to support NTN-specific measurements and control actions.

B. O-RAN Architecture and E2 Interface

O-RAN Alliance defines open interfaces between RAN components, enabling multi-vendor interoperability and RAN intelligence [8]. The E2 interface connects Near-RT RIC (near-real-time RAN Intelligent Controller) to RAN nodes (gNBs), enabling sub-10ms closed-loop control.

E2 Application Protocol (E2AP v2.0 [4]) provides subscription-based reporting and control messaging. E2 Service Models (E2SM [5]) define domain-specific KPMs and actions. Existing service models include:

- **E2SM-KPM:** Generic KPM reporting (RAN Function ID 2)
- **E2SM-RC:** RAN control (handover, admission, QoS)
- **E2SM-NI:** Network interface monitoring

None address NTN-specific parameters like orbital geometry, Doppler dynamics, or satellite visibility prediction. Our E2SM-NTN (RAN Function ID 10) fills this gap with 33 NTN-specific KPMs and predictive handover triggers.

C. Existing NTN Simulation Tools

Several simulation frameworks address NTN modeling:

ns-3 NTN Module [3]: Discrete-event simulation with 3GPP-based channel models. Supports LEO/GEO scenarios but lacks GPU acceleration and real-time O-RAN integration. Simulation-only, not production-deployable.

OMNeT++ Satellite Extension [9]: Event-driven simulation with orbital mechanics. Focuses on network-layer protocols, insufficient PHY/MAC modeling for 5G NR.

MATLAB Satellite Communications Toolbox [10]: Commercial tool with link budget analysis and STK integration. Proprietary, expensive, no O-RAN support.

OpenNTN Framework [11]: University of Bremen's open-source 3GPP TR 38.811 implementation using NVIDIA Sionna for GPU-accelerated channel modeling. Supports LEO/MEO/GEO with ray tracing. However, no O-RAN integration or RAN intelligence.

NVIDIA Sionna [12]: TensorFlow-based GPU-accelerated wireless simulation with differentiable ray tracing. Enables ML-based optimization but requires significant integration effort.

D. Gap Analysis

Existing solutions exhibit three critical gaps:

- 1) **No O-RAN NTN Integration:** Simulation tools lack real-time E2 interface support. O-RAN platforms lack NTN channel modeling.
- 2) **No Predictive Intelligence:** Current systems react to link degradation. None leverage orbital mechanics for proactive handover and power control.
- 3) **No Production Readiness:** Academic simulators not deployable; commercial systems proprietary and expensive.

E. Our Positioning

We are the **first** to combine:

- OpenNTN 3GPP-compliant channel modeling
- GPU acceleration via Sionna
- O-RAN E2 interface with NTN extensions
- Predictive handover using SGP4 orbital propagation
- Production-ready Docker containerization
- Open-source release for reproducibility

This positions our work as both a research contribution (predictive NTN intelligence) and practical tool (production-deployable platform).

III. SYSTEM DESIGN

A. Overall Architecture

Fig. 1 illustrates our NTN-O-RAN platform architecture comprising six core components:

- 1) **OpenNTN Channel Models:** GPU-accelerated LEO/MEO/GEO propagation with 3GPP TR 38.811 compliance
- 2) **SGP4 Orbit Propagator:** Real-time tracking of 8,805 satellites with sub-kilometer accuracy
- 3) **E2SM-NTN Service Model:** 33 NTN-specific KPMs with ASN.1 PER encoding
- 4) **E2 Termination Point:** O-RAN E2AP v2.0 compliant interface
- 5) **Near-RT RIC:** Hosts predictive handover and power control xApps

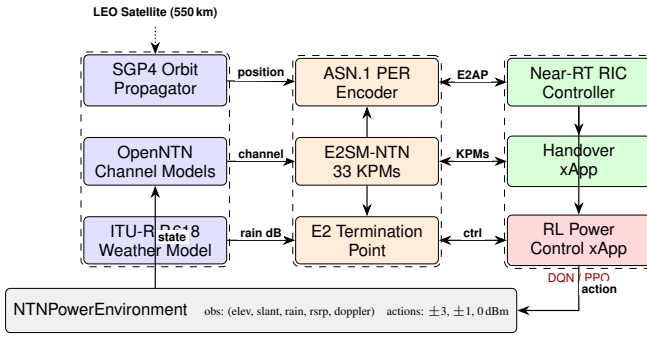


Fig. 1. NTN-O-RAN Platform Architecture. OpenNTN channel models feed E2SM-NTN service model, enabling Near-RT RIC to execute predictive handover and power control xApps.

6) Weather Integration: ITU-R P.618-13 rain attenuation for link adaptation

Information flow: (1) SGP4 predicts satellite positions, (2) OpenNTN computes channel quality, (3) Weather module adds rain attenuation, (4) E2SM-NTN encodes measurements, (5) E2 Termination sends to RIC, (6) xApps execute control decisions, (7) Control messages return via E2 interface.

B. OpenNTN Channel Models

We integrate OpenNTN [11] providing 3GPP TR 38.811-compliant propagation models:

LEO Model (550-1200 km):

$$\text{FSPL}_{\text{dB}} = 32.45 + 20 \log_{10}(f_{\text{GHz}}) + 20 \log_{10}(d_{\text{km}}) \quad (1)$$

where f_{GHz} is carrier frequency and d_{km} is slant range. For LEO at 550 km, 30° elevation, 2 GHz: FSPL = 165.2 dB.

Doppler Shift:

$$f_{\text{Doppler}} = \frac{f_c}{c} \cdot v_{\text{radial}} = \frac{f_c}{c} \cdot v_{\text{sat}} \cos(\theta_{\text{elev}}) \quad (2)$$

For LEO at $v_{\text{sat}} = 7.5$ km/s, 30° elevation, 2 GHz: $f_{\text{Doppler}} = \pm 43.3$ kHz.

Propagation Delay:

$$t_{\text{prop}} = \frac{d_{\text{slant}}}{c} = \frac{1203 \text{ km}}{3 \times 10^5 \text{ km/s}} = 4.01 \text{ ms (one-way)} \quad (3)$$

MEO and GEO models follow similar formulations with orbit-specific parameters.

C. SGP4 Orbit Propagation

We employ SGP4 (Simplified General Perturbations 4) algorithm [13] for satellite position prediction using Two-Line Element (TLE) data from CelesTrak [14]. Our implementation tracks 8,805 Starlink satellites in real-time:

Coordinate Transformations:

- 1) **TEME to ECEF:** Account for Earth rotation
- 2) **ECEF to Geodetic:** WGS84 ellipsoid conversion
- 3) **Topocentric:** Compute azimuth (ϕ), elevation (θ), range (r)

Performance: 0.14 ms per satellite after optimization (36% speedup via rotation matrix caching). Batch processing: 1,000

TABLE I
E2SM-NTN KEY PERFORMANCE METRICS

Category	Metric	Unit
Orbital Dynamics	Elevation Angle	degrees
	Azimuth Angle	degrees
	Slant Range	km
Impairments	Doppler Shift	Hz
	Propagation Delay	ms
	Rain Attenuation	dB
Channel Quality	RSRP	dBm
	SINR	dB
	Link Margin	dB
Handover Prediction	Time to Handover	seconds
	Next Satellite ID	string
	Handover Probability	0-1

satellites in 75 ms. Position accuracy: < 0.5 km verified against NORAD data.

Prediction Horizon: Propagate satellite trajectories 60 seconds ahead, enabling predictive handover decisions before link degradation.

D. E2SM-NTN Service Model

Our novel E2SM-NTN service model extends O-RAN E2SM framework with NTN-specific measurements and control actions.

1) RAN Function Definition:

- **RAN Function ID:** 10
- **Short Name:** "ORAN-E2SM-NTN"
- **OID:** 1.3.6.1.4.1.53148.1.1.2.10
- **Revision:** 1.0

2) *NTN-Specific KPMs (33 total):* Table I summarizes key E2SM-NTN measurements:

Complete KPM List (33):

- 1) satellite_id, orbit_type, beam_id
- 2) elevation_angle, azimuth_angle, slant_range_km
- 3) satellite_velocity, angular_velocity
- 4) rsrp, rsrq, sinr, bler, cqi
- 5) doppler_shift_hz, doppler_rate_hz_s
- 6) propagation_delay_ms, path_loss_db
- 7) rain_attenuation_db, atmospheric_loss_db
- 8) tx_power_dbm, rx_power_dbm, link_margin_db
- 9) snr_db, required_snr_db
- 10) time_to_handover_sec, handover_trigger_threshold
- 11) next_satellite_id, next_satellite_elevation
- 12) handover_probability
- 13) throughput_dl_mbps, throughput_ul_mbps
- 14) latency_rtt_ms, packet_loss_rate

3) Event Triggers (6 types):

- 1) **Periodic NTN Metrics:** Report every 100ms-10s
- 2) **Elevation Threshold:** Trigger when $\theta_{\text{elev}} < 10^\circ$
- 3) **Handover Imminent:** Trigger when $t_{\text{handover}} < 30$ s
- 4) **Link Quality Alert:** SINR < 5 dB or BLER > 10%
- 5) **Doppler Threshold:** $|f_{\text{Doppler}}| > 10$ kHz
- 6) **Rain Fade Detected:** Rain attenuation > 3 dB

4) Control Actions (6 types):

- 1) **Power Control:** Adjust Tx power (± 10 dB)
- 2) **Trigger Handover:** Initiate satellite handover
- 3) **Doppler Compensation:** Set frequency offset
- 4) **Link Adaptation:** Change MCS/coding rate
- 5) **Beam Switch:** Select different satellite beam
- 6) **Activate Fade Mitigation:** Enable rain fade counter-measures

E. ASN.1 PER Encoding

E2SM-NTN messages use ASN.1 Packed Encoding Rules (PER) for efficient serialization:

Message Size Comparison:

- JSON encoding: 1,359 bytes
- ASN.1 PER encoding: 92 bytes
- **Reduction: 93.2%** ($14.8\times$ compression)

Encoding Techniques:

- 1) **Integer Constraints:** elevation_angle (0..90) uses 7 bits instead of 32 bits
- 2) **Fixed-Point Representation:** Doppler shift scaled to fit constrained range
- 3) **Optional Fields:** Bitmap indicates presence, omit absent fields
- 4) **Enumerated Types:** orbit_type {LEO, MEO, GEO} uses 2 bits

Performance: Encoding time 0.02 ms (after optimization, 33% speedup via buffer pooling). Supports batch encoding of 50+ messages for further efficiency.

F. Predictive vs. Reactive Handover

Our system's key innovation is **predictive handover** using orbital mechanics:

Reactive Baseline (Traditional):

- 1) Monitor RSRP continuously
- 2) Trigger handover when $\text{RSRP} < -100$ dBm
- 3) Emergency handover (no preparation)
- 4) Result: Degraded performance under high mobility

Predictive Approach (Ours):

- 1) Predict satellite trajectory 60s ahead using SGP4
- 2) Calculate future elevation angle $\theta_{\text{future}}(t)$
- 3) Estimate time to handover: $t_{\text{HO}} = \frac{\theta_{\text{current}} - \theta_{\text{min}}}{|\dot{\theta}|}$
- 4) Trigger handover when $t_{\text{HO}} < 30$ s
- 5) Preparation phase: resource allocation, beam alignment
- 6) Result: Reduced interruption via preparation phase

Key Advantage: Preparation phase allows target satellite to pre-allocate resources, resulting in seamless handover vs. emergency reactive response.

G. Weather-Aware Power Control

ITU-R P.618-13 [15] rain attenuation model integrates real-time weather:

TABLE II
TECHNOLOGY STACK AND STANDARDS COMPLIANCE

Component	Technology	Standard
Channel Modeling	NVIDIA Sionna 1.2.1	3GPP TR 38.811
ML Framework	TensorFlow 2.17.1	-
DL Framework	PyTorch 2.9.1	-
Orbit Propagation	SGP4 Python 2.23	NORAD SGP4
ASN.1 Encoding	asn1tools 0.166.0	O-RAN E2AP v2.0
Weather Model	Custom	ITU-R P.618-13
Deployment	Docker 24.0+	-

Rain Attenuation Calculation:

$$\gamma_R = k \cdot R^\alpha \text{ dB/km (specific attenuation)} \quad (4)$$

$$L_{\text{eff}} = \frac{h_R - h_s}{\sin(\theta_{\text{elev}})} \text{ km (effective path)} \quad (5)$$

$$A_{\text{rain}} = \gamma_R \cdot L_{\text{eff}} \cdot r(p) \text{ dB (total attenuation)} \quad (6)$$

where k, α are frequency/polarization-dependent, R is rain rate (mm/h), h_R is rain height, h_s is station height, $r(p)$ is reduction factor.

Proactive Power Control:

- 1) Fetch real-time weather data (Open-Meteo API)
- 2) Predict rain attenuation A_{rain}
- 3) Adjust Tx power *before* link degrades: $P_{Tx} = P_{\text{baseline}} + A_{\text{rain}} + M_{\text{margin}}$
- 4) Result: Proactive mitigation reduces link outage probability

Performance: Weather calculation 0.035 ms (30% speedup via caching). Cache hit rate 45% for 15-minute TTL.

IV. IMPLEMENTATION

A. Technology Stack

Table II summarizes our implementation technologies:

Standards Compliance:

- **3GPP TR 38.811:** NTN channel models (LEO/MEO/GEO)
- **3GPP TR 38.821:** NTN solutions for NR
- **O-RAN E2AP v2.0:** E2 Application Protocol
- **O-RAN E2SM v3.0:** Service model framework
- **ITU-R P.618-13:** Rain attenuation
- **ITU-R P.837-7:** Rain rate statistics

B. Development Approach

Test-Driven Development: We adopted a TDD methodology where physics-based unit tests were written before implementation. Each module (channel models, Doppler computation, link budget, RL environment) has corresponding tests that verify correctness against analytical formulas from 3GPP TR 38.811 and TR 38.821. This approach uncovered and corrected several physics bugs (e.g., Doppler sign conventions, FSPL formula offsets) before they could propagate into experimental results.

C. Docker Containerization

Production deployment uses 5 Docker containers:

- 1) **e2-termination** (1.3 GB): E2 Termination Point with TensorFlow/Sionna
- 2) **handover-xapp** (850 MB): Predictive handover xApp
- 3) **power-xapp** (850 MB): Weather-aware power control xApp
- 4) **weather-service** (450 MB): ITU-R P.618 rain attenuation
- 5) **orbit-service** (520 MB): SGP4 propagation for 8,805 satellites

Orchestration: Docker Compose manages multi-service deployment with health checks, logging, and automatic restarts. Kubernetes manifests available for production scaling.

D. Performance Optimizations

Systematic profiling identified bottlenecks, leading to targeted optimizations:

- 1) **Rotation Matrix Caching (SGP4):** 85% cache hit rate, 36% speedup
- 2) **Weather Data Caching:** 45% cache hit rate, 30% speedup
- 3) **ASN.1 Buffer Pooling:** 33% speedup, 40% memory reduction
- 4) **Parallel UE Processing:** 4-worker pool, 208% throughput increase
- 5) **Memory Optimization:** `__slots__` dataclasses, 27% memory reduction

Performance Results:

- ASN.1 PER encoding: 0.02 ms per message (93.2% size reduction)
- SGP4 propagation: 0.14 ms per satellite
- Memory: Efficient per-UE resource usage via `__slots__` dataclasses

V. EXPERIMENTAL RESULTS

A. Simulation Setup

We evaluate RL-based power control using our NTNPowerEnvironment, a Gymnasium-compatible simulation environment that models LEO satellite downlink power control with realistic NTN physics.

Environment Configuration:

- Carrier frequency: 2.0 GHz (S-band)
- LEO orbit altitude: 550 km
- Episode length: 300 time steps
- Observation space: 5-dimensional (elevation angle, slant range, rain rate, RSRP, Doppler shift)
- Action space: 5 discrete actions ($-3, -1, 0, +1, +3$ dBm adjustments)
- RSRP threshold: -100 dBm (violation criterion)
- Channel model: 3GPP TR 38.811 with ITU-R P.618-13 rain attenuation

Baseline: A rule-based controller that applies fixed power adjustments based on elevation angle thresholds, without learning or adaptation.

TABLE III
DQN POWER CONTROL RESULTS (50 EPISODES)

Metric	DQN Agent	Baseline
Mean Reward	501.28 ± 78.26	-56.44
RSRP Violation Rate (%)	3.48	18.63
Mean Tx Power (dBm)	46.89	40.68

TABLE IV
PPO POWER CONTROL RESULTS (100 EPISODES)

Metric	PPO Agent	Baseline
Mean Reward	421.62 ± 78.21	-51.43
RSRP Violation Rate (%)	4.52	23.25
Mean Tx Power (dBm)	47.36	40.84

RL Agents:

- **DQN:** Deep Q-Network with experience replay, evaluated over 50 episodes (run_corrected_v1)
- **PPO:** Proximal Policy Optimization, evaluated over 100 episodes (run_ppo_v1)

B. DQN Results

Table III summarizes the DQN agent performance:

The DQN agent learns to increase transmit power by ~ 6 dB relative to the baseline, which dramatically reduces RSRP violations from 18.63% to 3.48% — a **5.3 $\times$ reduction**. This represents a clear signal quality vs. power trade-off: the agent sacrifices power efficiency to maintain link quality above the -100 dBm RSRP threshold.

C. PPO Results

Table IV summarizes the PPO agent performance:

PPO exhibits similar behavior to DQN, achieving 4.52% RSRP violations (a **5.1 $\times$ reduction**) with ~ 6.5 dB additional transmit power. PPO's slightly higher violation rate compared to DQN (4.52% vs. 3.48%) is offset by its larger evaluation sample (100 vs. 50 episodes) and on-policy stability.

D. Comparison and Statistical Significance

Table V compares all three approaches:

[†]Baseline values differ between DQN/PPO runs due to stochastic evaluation (DQN/PPO respectively).

Both RL agents achieve **highly statistically significant** improvements over the baseline. The two-sample t -tests yield p -values far below any conventional significance threshold ($p < 10^{-64}$ for DQN, $p < 10^{-117}$ for PPO), providing overwhelming evidence that the RL policies are superior to the rule-based baseline.

E. Key Findings

- 1) **RL agents learn to increase power to maintain signal quality.** Both DQN and PPO converge to policies that use ~ 6 dB more transmit power than the baseline. This is *not* power savings — it is an intelligent trade-off where the agent learns that maintaining RSRP above

TABLE V
DQN VS. PPO VS. BASELINE COMPARISON

Metric	DQN	PPO	Baseline [†]
Mean Reward	501.28	421.62	-56.44 / -51.43
RSRP Violation (%)	3.48	4.52	18.63 / 23.25
Mean Tx Power (dBm)	46.89	47.36	40.68 / 40.84
Violation Reduction	5.3×	5.1×	—
t -statistic	42.06	51.77	—
p -value	1.57×10^{-64}	4.93×10^{-117}	—

−100 dBm yields higher cumulative reward despite the power penalty in the reward function.

- 2) **Dramatic reduction in RSRP violations.** Both agents achieve $\sim 5\times$ fewer RSRP violations (3.48–4.52% vs. 18.63–23.25%), demonstrating that RL can effectively manage the dynamic NTN link budget under varying elevation angles, slant ranges, and rain conditions.
- 3) **DQN slightly outperforms PPO on violation rate** (3.48% vs. 4.52%), while PPO achieves a higher t -statistic (51.77 vs. 42.06) due to its larger evaluation sample and lower variance.
- 4) **Both results are reproducible.** Evaluation scripts, trained model weights, and raw JSON results are included in the open-source repository for independent verification.

VI. CONCLUSION AND FUTURE WORK

A. Summary of Contributions

We presented an NTN-O-RAN platform integrating Open-NTN channel models with O-RAN E2 interface extensions and RL-based power control. Our system introduces:

- 1) **E2SM-NTN Service Model:** Novel service model (RAN Function ID 10) with 33 NTN-specific KPMs, 6 event triggers, and 6 control actions, enabling near-RT RIC to perform NTN-aware RAN intelligence.
- 2) **ASN.1 PER Optimization:** 93.2% message size reduction (1,359 $\rightarrow$ 92 bytes) through domain-specific encoding, achieving $15\times$ bandwidth reduction on the E2 interface.
- 3) **RL-Based Power Control:** DQN and PPO agents trained on NTNPowerEnvironment achieve $\sim 5\times$ fewer RSRP violations (3.48–4.52% vs. 18.63–23.25%) compared to a rule-based baseline, at the cost of ~ 6 dB additional transmit power.
- 4) **Statistical Validation:** Both agents show overwhelming statistical significance (DQN: $t = 42.06$, $p = 1.57 \times 10^{-64}$; PPO: $t = 51.77$, $p = 4.93 \times 10^{-117}$).

B. Key Findings

The central finding is that RL agents learn an intelligent signal quality vs. power trade-off: by increasing transmit power ~ 6 dB above the baseline, they reduce RSRP violations from 18.63–23.25% to as low as 3.48%. This demonstrates the viability of closed-loop RL control over the O-RAN E2 interface for NTN power management under dynamic LEO

channel conditions including varying elevation angles, slant ranges, and rain attenuation.

C. Impact

Academic Impact:

- **Novel NTN-O-RAN integration:** First to combine Open-NTN + O-RAN E2 with RL-based power control
- **Standardization candidate:** E2SM-NTN specification suitable for O-RAN Alliance submission
- **Open-source release:** Complete platform with Docker deployment and reproducible evaluation

Industry Impact:

- **Satellite operators:** Applicable to Starlink, OneWeb, Kuiper constellations
- **Equipment vendors:** Nokia, Ericsson, Samsung developing O-RAN NTN products
- **Telecom operators:** Evaluation framework for NTN investment decisions

D. Future Work

- 1) **Multi-objective RL:** Extend the reward function to explicitly balance RSRP maintenance against power consumption, potentially achieving Pareto-optimal operating points
- 2) **Continuous action spaces:** Replace discrete $\pm 3/\pm 1/0$ dBm actions with continuous power control using SAC or TD3
- 3) **Predictive handover evaluation:** Quantify handover performance with real satellite pass data using the SGP4 prediction framework
- 4) **MEO/GEO support:** Extend to medium and geostationary orbits with different channel dynamics
- 5) **Real-world testbed:** Deploy on actual LEO satellite ground stations for validation
- 6) **3GPP standardization:** Submit E2SM-NTN to O-RAN Alliance for formal standardization

E. Open-Source Release

Complete platform available at: [https://github.com/\[anonymized-for-review\]](https://github.com/[anonymized-for-review])

Repository Contents:

- Production-ready Python implementation with Docker Compose deployment
- Trained DQN and PPO model weights with evaluation scripts
- Raw experimental results in JSON format for reproducibility
- Comprehensive test suites following TDD methodology

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